



MSc Advanced Computing

Version	1				
Course Type	Postgraduate				
Award Type	Master of Science - MSc				
Code	0734	SCQF Level	Level 11	QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	180	ECTS Credit Points	90		
Valid From	2026 September (September 2026) as of (September 2026)				
School	School of Computing, Engineering and Technology				

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100366	computer science	computer science	100%
			Total 100%

What the Course Involves	The general format of the first two semesters for full-time students comprises a mix of lectures and lab sessions. Students will have access to all materials on the University's VLE. In semester 3, students are engaged on a project module, and have a structured timetable as well as meetings with their project supervisor to discuss progress and offer support. Part-time students attend the same programmed sessions as the full-time students but over an extended duration corresponding to two 15-credit modules per semester.
	Full-Time Students - For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) - For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January) Part-Time Students - For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) - For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters)

Online Learning students will not have face-to-face sessions. Instead, they will have access to lecture, tutorial and laboratory electronic (video, audio, written) materials on the University's online learning platform Moodle, together with online support sessions. The module schedule is the same as that for on-campus students (full-time and part-time).

Course Collaborations

No collaborations listed

Introduction to the Course

The MSc/PgDip/PgCert Advanced Computing is designed for graduates from disciplines with significant computing content who wish to extend their knowledge across a broad range of contemporary computing topics. The course provides advanced study in key areas of modern computing, including scalable systems development, cyber security, data technologies, and artificial intelligence. Elective modules allow students to tailor their learning towards areas of particular interest, such as cutting-edge computing research, or further advanced topics in cyber security, data science, games design and development, and software development.

The MSc project module is of particular value to employers as evidence of the student's ability to produce well-documented, high-quality and deadline-driven work of an advanced technical and presentational standard.

At all stages of study, the course content is complemented by practical work intended not only to reinforce the learning process, but also to enhance the vocational relevance of the course of study. The syllabus content incorporates course modules intended to enhance the technical and professional skills required to design, develop and evaluate state-of-the-art computational systems.

The RGU course is designed to foster the development of individuals into evidence-based, critical thinkers. It aims to enhance the acquisition of knowledge, skills, and behaviours that are crucial for their prospective employers, irrespective of the subject area. The course curriculum is shaped by pertinent local, national, and international policies, frameworks, and strategies, ensuring that the content is both current and relevant. The main themes in the course include, critical thinking and writing, evidence, research and critical appraisal, data management and presentation, focused project work and subject specific modules. The module design enhances their relevance and the practical application of theory to real-world issues and scenarios. This integration of theory into the students' professional environments supports their development. Furthermore, RGUplus (2023) will offer opportunities that empower students to adopt a global perspective, thereby enhancing their employability and providing them with the necessary skills and knowledge to excel after university. RGUplus represents the collective offerings of opportunities, experiences, and learning provided at RGU based on the three pillars of Sustainability, Entrepreneurial Mindset, and Learning Beyond the Classroom.

Module assessments range from a portfolio of evidence, essays, timed assessments and written reports depending on the course taken. Formative assessments form a key part of learning in all modules and activities are constructively aligned towards the final module assessment. RGU's dedication to sustainability recognises that every course, subject area, and professional discipline is connected to the global sustainability agenda and plays a crucial role in making a positive contribution to the United Nations Sustainable Development Goals (United Nations 2024). Consequently, every course curriculum integrates content on aspects of sustainability that are deemed relevant to the subject area. Additionally, the principles of self-assessment, skill building, and the development of intentional careers, form a crucial foundation to ensure that graduates can maximise their opportunities for future employability globally.

Online students will, after an immersive online face-to-face induction, engage with the learning using the online learning platform Moodle. They will experience a variety of learning methods such as synchronous and asynchronous lectures, tutorials and study groups, as well as various forums for discussion and debate.

The specific aims for each stage of the course are:

Educational Aims of the Course	Postgraduate Certificate To produce practitioners who will be able to: • Appraise modern techniques in advanced computing. • Construct solutions using appropriate advanced computing technologies. • Analyse contemporary challenges in advanced computing and appropriate technological approaches to address them. Postgraduate Diploma To produce practitioners who, in addition to the above, will be able to: • Appraise techniques for constructing scalable software. • Evaluate methods for knowledge representation, and subsequent reasoning therewith. • Analyse the influence of human behaviour and usability on the design and effectiveness of secure systems. • Evaluate the legal, ethical and societal implications of the development and use of artificial intelligence technologies. Master of Science To produce practitioners who, in addition to the above, will be able to: • Appraise the requirements of a substantial advanced computing problem. • Analyse different design approaches and technical solutions for an advanced computing project. • Take responsibility for a substantial advanced computing project. • Construct a solution involving different advanced computing technologies. • Appraise a complex advanced computing solution. • Take responsibility for writing and presenting to an intended audience.
Awarding Institution	Robert Gordon University
Course Duration	On Campus Full-Time Students Full-time 12 months with a September intake Full-time 16 months with a January intake On Campus Part-Time Students Part-time 28 months with a September intake Part-time 32 months with a January intake Online Learning Full-Time Students

	Full-time 12 months with a September intake Full-time 16 months with a January intake Online Learning Part-Time Students Part-time 28 months with a September intake Part-time 32 months with a January intake Depending on their mode of study, students can choose to undertake either a short-term (3 months) or long-term (12 months) placement. All placements require approval from the School prior to commencement. If a student goes on a placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course offers advanced technical skills across key areas of contemporary computing, including scalable software systems, cyber security, data technologies, and artificial intelligence. Alongside the core modules, a range of elective options allows students to tailor their learning towards areas of particular interest. These subjects align with the School's strengths across computing research and applied development, meaning students will benefit from being taught by academics who are active in research and industry-facing innovation across these domains. The course will benefit from collaborations with organisations such as The Data Lab and industry partners, and will also gain from inter-university links developed through existing research collaborations. If applicable, the long placement of up to 45 weeks offers students the opportunity to develop computing skills and build teamwork and networking skills in a realistic business environment. There is also a 4–20 week short placement opportunity to develop the student's professional and technical skills. Throughout the course, students will engage with contemporary computing technologies from both theoretical and practical perspectives. This includes the design and development of secure and scalable systems, the evaluation of emerging technologies, and consideration of the professional, legal and ethical issues associated with modern computing practice.
	Students may elect to undertake a short placement or a year-long placement. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor.

Placement/Professional Experience	As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	The course has a strong grounding in research techniques and enables students to advance into research and development at doctoral level within this institution or other UK higher education institutions.
External and Internal Indicators of Quality and Standards	SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide the student with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate programme. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Postgraduate Programme Management Team, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education

	Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	The normal entry requirements are an Honours Degree (2nd class or better) in a discipline with significant Computing content. Applicants with a lesser qualification who possess significant industrial experience of substantial relevant personal development in a relevant field may be considered for entry into the course.
Alternative Qualifications for Entry	Where an applicant supports an application with alternative qualifications, the School may wish to arrange an interview to discuss the level of qualification offered and the experience gained in an industrial background.
Interview Procedures	Interviews are not normally required.
Course-Specific Regulations	The MSc project module cannot be condoned as it is key to demonstrating essential skills in the course, such as the ability to conduct a significant piece of independent work. The placement modules cannot be used to compensate for failed modules. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.

Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.

Course-Specific Learning Experience

The extent of the course syllabus demands a relatively rapid pace of delivery. Consequently, student-centred learning strategies play an important role in most subjects and throughout all stages, reinforcing and supplementing more traditional methods of subject delivery.

Directed reading, independent progress through computer-based exercises, independent reading, analysis of case studies and directed searches of on-line information resources are all commonplace.

The preferred strategy for any course module depends on the nature and extent of the syllabus content, on its subject-specific learning outcomes and its contribution to the overall course aims. In the sections that follow, a more detailed description of the learning experience is given, which distinguishes between:

- Core technological modules
- Professional skills modules
- Integrative course modules

Teaching and learning are reviewed through an annual appraisal mechanism and strategies are liable to change for the sake of continuing improvement and to reflect best practice.

Part-time students participate in the same sessions as full-time students, but they take 2 modules (30 credits) per semester instead of 4 modules (60 credits).

Core Technological Modules

These modules impart specialist technical knowledge through the medium of lectures and tutorials, supplemented as necessary by student-centred materials and tasks. All core technological modules have a supporting practical element, which is delivered through a series of exercises that are designed to foster analytical skills and to illustrate the synthesis of fundamental knowledge and problem solving techniques in a structured problem domain.

For online learning students, lectures will be provided asynchronously (recorded) and tutorials will be online, supplemented by dedicated face-to-face online support sessions.

Professional Skills Modules

Technical report writing skills are exercised in all modules. The course addresses project planning skills that are vocationally relevant and that also directly underpin the planning and management of the MSc Project. Consideration of legal, ethical and professional issues is embedded throughout all course modules. Student feedback in this course takes the form of critical self-appraisal, peer appraisal, and commentary from academic staff across all modules. Through directed reading and by listening to invited external speakers, students are also encouraged to develop a broader perspective of advanced computing topics and to sharpen their awareness of the need for professional standards and ethics in the workplace.

Recorded seminars will be made available to distance learning students where possible.

Integrative Course Modules

The MSc Project module serves the primary purpose of integrating technological and professional strands, which are developed in the preceding PgDip stage, and do so in the context of a substantial advanced computing project. Students are required to investigate a technical problem and its domain, examine relevant methods and tools to gain some understanding of the problem, design a specification for solving the problem, then develop and evaluate that solution. The problem is original to the student and its solution therefore requires innovative application of knowledge and techniques either studied throughout their course, or acquired through independent research of recent and relevant literature.

The MSc Project provides a vehicle for integrating specialist knowledge with analytic, problem solving, managerial and communication skills.

Research Skills

	The course framework underpins the MSc project in a number of ways. The core technological modules provide skills in specific subject areas that will feed into the theoretical and technical development of the project. The professional skills modules will provide students with: • Core transferrable skills in how to carry out research in a computing domains, allowing them to place their project in context through a thorough exploration of relevant literature. • Knowledge of cutting-edge developments in AI, which will ensure their project has importance and relevance in a fast-moving subject area. • An understanding of ethical, legal, professional and societal considerations around the use of AI, allowing them to evaluate their project in this context towards ensuring responsibility and sustainability. All students who are about to embark on the MSc Project are encouraged to attend an Library Workshop. This is presented by staff from the Library Service and is intended to familiarise students with available information sources (journals, monographs and on-line resources) and to inform them of systematic and effective methods to review current literature in the field of computing. The Workshop is not assessed directly but evidence of independent research and a review of the project context are elements of the overall assessment of the MSc Project. <u>Group Work</u> Group exercises are used to develop team-working skills and encourage discussion. This type of work will be used in some technological core modules as well as in the professional skills modules. Appropriate technologies and pedagogical methods will be used to support group working.
	<u>Design Philosophy</u> The course curriculum is designed to enable the student to acquire and develop their subject knowledge and understanding, intellectual and thinking ability and practical and key transferable skills at Masters level corresponding to the Scottish Credit and Qualifications Framework. As students progress through the course, they are encouraged to expand their understanding and critical appreciation of topics in advanced computing, including underlying theories, applications, and ethical foundations. The overall philosophy and strategy for the course is to provide a sound knowledge and understanding of key topics in advanced computing, in terms of both underlying theory and the broader contexts in which it is and may be used. The vocationally orientated nature of the course requires the student to achieve general and specific skills, which are addressed through a combination of teaching and learning methods, including lectures, lab sessions, tutorials, seminars, student centred learning and private study. For distance learning students, on-line materials (some of them interactive) and on-line sessions will replace face-to-face sessions and seminar/guest lecture presentations may be streamed and/or recorded and uploaded to the Virtual Learning Environment. The course induction programme is aimed at ensuring students gain familiarity with the (academic and non-academic) policies and procedures required for the successful completion of the course, e.g. assessment, extenuating circumstances, grading schemes, employability, student services etc. Students are expected to take an active role in their learning. As well as studying formally delivered materials, they are required to expand their knowledge and skills by reading related materials, and undertaking increasingly more challenging theory- and practice-based research. As the course progresses, students will be expected to self learn, work independently, critically reflect on their learning and establish their own learning objectives. Students are also expected to, where appropriate, seek help from Module Coordinators and others as appropriate. <u>Learning and Teaching Methods</u>

Teaching and Learning Strategy

The main goal of the teaching and learning strategy is to enable students to become critical, reflective, independent learners who are active participants in the learning process. Student involvement in the learning process ensures the development of interpersonal skills and professional attitudes as well as developing the motivation and autonomy needed to take control and regulate their own educational experiences. This means that, throughout their course of study, the student will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their course, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide range of appropriate delivery styles. These include conventional instructional teaching methodologies, social learning activities such as group and team-working, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

As well as the day-to-day use of the latest computer hardware and software afforded by the degree, extensive use of technology-assisted learning is made within the course of study. Examples of these include assessment by means of computer-assisted assessment packages embedded in the virtual learning environment to engage the student and increase the speed of feedback and the use of an integrated approach whereby the student has short periods of instruction followed by immediate lab exercises to contextualise the topic.

The broad nature of the course including common and specialist elements necessitates the use of a broad range of teaching techniques. These include lectures, seminars, tutorials, computer laboratories, interactive materials, distance learning online supported sessions, group work and the undertaking of a large supervised project.

Contact Time

For on-campus students the pattern of contact will involve four timetabled classes per week. While individual modules may have bespoke arrangements to suit their topic, in general students can expect a weekly structure per module of:

- 1 hour lecture or seminar
- 2 hours practical laboratory time
- Up to 2 hours of staff "office hours" for additional support.

For online students, the pattern of contact will involve asynchronous, recorded lectures, online seminars and tutorials, and dedicated help and support sessions. While individual modules may have bespoke arrangements to suit their topic, in general students can expect a weekly structure per module of:

- 1 hour recorded lecture or seminar
- Dedicated online support session for completing practical work
- Up to 2 hours of staff "office hours" for additional support

Additionally for online students, certain modules will include opportunities for synchronous online tutorials and group discussions lasting 1-2 hours each.

Lecture

This is usually a formal discourse for the purposes of disseminating information, the demonstration of techniques, and (where relevant) the discussion of supporting ideas and consequences. The lecture is supported by a full range of dissemination equipment, including whiteboard, video, and computer projection facilities where appropriate. For online students, lectures will be delivered asynchronously as recordings.

Online Seminar and Tutorials

In the School of Computing, Engineering and Technology, seminars and tutorials are used for a wide range of activities, each suited to the particular subject. Some tutorials consist of a member of staff helping small groups of students work through problems and exercises, while others involve group exercises, where each group is encouraged to discuss a problem, plan a solution, before presenting this to the rest of the tutorial session. Generally, this type of teaching is used to support the lecture as a way of clarifying the content, and expanding and consolidating knowledge. For online students, breakout rooms and screen-sharing are used to encourage discussion and collaboration.

Online Support

Online materials, sometimes interactive, will be used to both explain theoretical aspects of modules and to guide students on practical exercises. Video, audio and text media will be used. Interactive online support sessions will be used to both explain theoretical aspects of modules and to guide distance learning students on practical exercises. Discussion sessions will be used to support a critical appraisal of modules.

Project

The MSc Project serves the primary purpose of integrating technological and research strands, which are developed in the preceding PgDip stage, and does so in the context of a substantial project in a relevant area of advanced computing.

The project typically involves the development and evaluation of the solution to a problem. The problem is original to the student and its solution therefore requires the innovative application of knowledge and techniques, either studied in the previous PgDip stage or acquired through independent research of recent and relevant literature.

The MSc Project provides a vehicle for integrating specialist knowledge with analytic, problem solving, managerial and communication skills.

Learning Resources

The course will make sure of a wide range of learning resources. Staff will upload module-specific course materials to the Virtual Learning Environment, including lecture materials, guided tutorial questions and solutions, guided laboratory work, case studies, coursework specifications and guidelines and sample solutions to exercises as appropriate. Feedback on performance will be given during tutorials and/or laboratories to enable the student to learn from past performance. Module coordinators are responsible for ensuring that students are aware of essential readings and additional textbook recommendations at the beginning of each module.

The library stocks a high number of learning resources with preference given to electronic resources. The library regularly searches for new materials and reviews its current offerings, updating its portfolio where appropriate. Academic staff also submit resource recommendations for consideration by the library.

Learning Environment

Students may contact a wide variety of staff depending on their needs:

- A Module Coordinator or any other members of staff involved in the delivery of the module: module related queries. Their Course Leader/Tutor: course queries and/or personal issues.
- The Postgraduate Administrative Assistant; non-academic matters related to the course.
- A member of the systems administration team: for technical support in the School's computer laboratories and/or for setting up software in their own devices.
- IT services: for other technical support.
- Library staff.
- Student services staff.
- Student Union.

	Students may use a variety of communication methods depending on their needs, including face-to-face, email, and Teams calls and chats. The School has a number of computer laboratories for exclusive use of computing students. These are state of the art facilities with up to date software. Supervised (online) practical sessions are provided for most modules, while remote access to laboratory PCs is available for online learning students. Furthermore, all students have remote access to a Virtual Desktop that replicates the on-campus laboratory environment, providing access to the same software, tools, and computing resources used in teaching. Moodle is a widely used virtual learning environment which not only enables academic staff to upload materials but also contains a wide variety of facilities including forums and a site for the submission of courseworks by students. <u>Student's Role in Learning</u> The student is expected to be an active participant in the learning process, becoming an active member of a learning community in which students learn from each other as well as from staff. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which may form the basis of tutorial guidance with staff. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library.
Assessment	A range of assessment methods are used, with particular emphasis placed on coursework assignments, including design exercises (ranging from design studies to fully implemented solutions), comparative studies, and investigative reports. Students are told in advance what assessment methods will be used for each module and will have access to examples of past assessments. Where the format of assessment varies significantly from one year to the next, examples of assessment will be created in order to ensure students understand what the assessment entails. All assessments include a marking/grading scheme which describes the criteria used for the award of marks/grades. Technical reports will be used to document development-oriented assignments. Structured project documents will specify the management of the project conduct, quality guidelines and project deliverables. It will also contain information on the use of web resources to present interim MSc Project progress updates and guidelines on the live presentation and demonstration of MSc Project outcomes. The use of IT tools will be assessed in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions. In certain assessments, students will be required to use generative AI tools (e.g. Copilot) to complete all or part of their submission. This will allow them to put into practice the technical, ethical and legal skills acquired in the course, towards effective and responsible use of AI in completing a defined task. In such assessments, considerable weighting will be placed on how the tools were used (e.g. approaches to prompt engineering) and the students' overall evaluation of the quality of the subsequent output.
	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies;

University Support for Teaching, Learning and Assessment	• the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The School's computing facilities comprises a number of labs with state-of-the-art PCs, each updated over a 3 year cycle to ensure they deliver high performance and support the latest software. Remote access to laboratory PCs allows online learning students to access the facilities from elsewhere. The School believes the academic success of the students is founded in the desire to develop and maintain close links with industry and to actively seek state-of the-art technologies and integrate these into course modules. All students have access to their Course Leader and the Postgraduate Administrative Assistant in the School who work as a team in order to ensure that comprehensive support is given to students not only on academic matters but also pastoral care. The project coordinator provides general advice in all aspects of projects. In addition, all students are assigned a project supervisor who provides detailed advice on the project. Students going on placement will also be supported by a placement supervisor at their place of work and an academic tutor for the placement. Moodle, the University's Virtual Learning Environment, is used for a variety of purposes which support learning including: (i) a repository for learning materials; (ii) a discussion forum; (iii) a place for the uploading of courseworks by students; (iv) a tool for electronic assessment; (v) a facility for feedback to students. In addition to the induction sessions offered at the start of the course, the School offers weekly focused induction sessions during Semester 1. Each session covers one or more topics of interest to students such as assessment, research, student services, academic misconduct and dealing with situations affecting performance.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups.

	Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	The School of Computing, Engineering and Technology is committed to the highest standards of quality, best pedagogy and research within an environment which values all associated stakeholders. Formal feedback is gathered from Student Questionnaires and informal feedback is gained from focus groups and informal chats with students. The student questionnaires provide feedback on the course and modules and give students the opportunity to comment in detail on all aspects of the course as well as the general environment, support services and facilities. The focus groups discuss a variety of issues in the course with an emphasis on issues raised by students. Responses to, and actions resulting from, these comments are addressed fully in an Annual Course/Programme Appraisal and widely discussed by the Postgraduate Management Team. Responses to issues raised at focus groups are given as soon as practically possible. In addition, the students have an opportunity to discuss with their Course Leader any course or management issues as and when required either in person or via their class representative.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.

Course-Specific Student Feedback Mechanisms

The University uses a variety of mechanisms to obtain feedback from students about their learning experience. The University has a system of student representation, via the appointment of Student Representatives for each course, who in turn represent the interests of their colleagues at Staff/Student Liaison meetings held each semester with staff representatives from the School concerned. Student Representatives are also invited to attend a meeting with the Dean of School during the first and second semester and with the University Principal in the second semester to discuss a selection of topics and to air any concerns they have. The notes of all these meetings and responses to any issues raised are subsequently made available to students.

A single Staff/Student Liaison Committee exists for all postgraduate courses in the School, which meets twice a semester. The Committee comprises the Dean of School, Course Leaders, teaching staff and class representatives from each course intake and meets at least once per semester to exchange information and discuss issues. While it is unlikely that online learning students will be able to participate "live" in the Committee, they will be asked to submit any issues in advance of the meeting and responses will be given to them after the Committee.

Awards**Stage 1**

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	Advanced Computing	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Advanced Computing	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Advanced Computing	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

Description	SCQF Characteristic	Soft Skill	Sub Soft Skill
Appraise state-of-the-art approaches to solving a broad range of problems in advanced computing.	SC1	Course Level	SCQF 11
Evaluate tools and techniques which aid the specification, design and implementation of an advanced computing project.	SC1	Course Level	SCQF 11
Make informed judgements about project management frameworks and lay down effective procedures for monitoring progress of any large scale research/development project.	SC1	Course Level	SCQF 11
Evaluate information resources and how to access these to obtain state-of-the-art knowledge of current advanced computing tools, theories and technologies.	SC1	Course Level	SCQF 11
Synthesise comprehensive, detailed, state-of-the-art knowledge of the specialist area(s) covered by the course within the context of the broader discipline or field within which the course is set.	SC1	Course Level	SCQF 11
Construct research methods appropriate specifically to their Masters independent study project, together with detailed knowledge of the particular area or sub-area in which the project is carried out.	SC1	Course Level	SCQF 11
Analyse the theoretical and empirical limits and boundaries of a variety of topics in advanced computing, and the range of methods of study and types of judgments employed by advanced practitioners.	SC1	Course Level	SCQF 11
Construct statistical methods to assess and evaluate research or experimental test results.	SC2	Course Level	SCQF 11
Deal with a wide range of theories, tools, techniques and skills related to advanced computing.	SC2	Course Level	SCQF 11
Produce high quality coursework in the PgCert/PgDip modules through effective use of available IT tools and packages.	SC2	Course Level	SCQF 11
Take responsibility for an advanced computing project, and to enhance the production and presentation of the Masters stage report and oral defence.	SC2	Course Level	SCQF 11
Analyse an advanced computing problem from a critical perspective, identifying key requirements, alternative solutions, and evaluation methods.	SC2	Course Level	SCQF 11, SCQF 11
Create an effective, original solution to an advanced computing problem.	SC3	Course Level	SCQF 11
Make informed judgements in situations where information is incorrect and/or incomplete.	SC3	Course Level	SCQF 11
Evaluate the efficacy of proposed solutions.	SC3	Course Level	SCQF 11
Evaluate the selection and application of analysis/test strategies for validating research outcomes, or testing a developed system and compliance to initial requirements.	SC3	Course Level	SCQF 11
Produce a proposal for the Masters project.	SC3	Course Level	SCQF 11
Generate the Masters project, applying an analytic, rigorous and critical approach to identifying project goals, proposing and evaluating different research/solution strategies, and generating a useful and effective output.	SC3	Course Level	SCQF 11
Construct effective communications, using appropriate methods to audiences with different level of knowledge and understanding.	SC4	Course Level	SCQF 11
Produce structured and coherent written reports to document the investigation and survey of a problem domain or an			

advanced computing task.	SC4	Course Level	SCQF 11
Construct strategy and design decisions, and record research or test strategies and results throughout the various stages of a major advanced computing project using appropriate software tools.	SC4	Course Level	SCQF 11
Produce a clear and coherent thesis which documents the process and end result of a major advanced computing exercise.	SC4	Course Level	SCQF 11
Analyse a variety of data, including numerical data.	SC4	Course Level	SCQF 11
Take responsibility for an oral defence of the work undertaken during the Masters project before a live audience of peers and/or academics.	SC4	Course Level	SCQF 11
Prepare concise and relevant reviews of work undertaken within a predefined plan of work, and make a valuable contribution to team work tasks.	SC4	Course Level	SCQF 11
Criticise specialist areas of research.	SC5	Course Level	SCQF 11
Take responsibility for working as part of a team, contributing effectively and taking significant responsibility for the work of the team.	SC5	Course Level	SCQF 11
Take responsibility for working independently, demonstrating a high level of autonomy and own initiative, taking responsibility for own work.	SC5	Course Level	SCQF 11
Produce effective communications, both orally and in writing.	SC5	Course Level	SCQF 11
Synthesise critical reflection into work roles and responsibilities.	SC5	Course Level	SCQF 11
Analyse tasks to identify problems, solutions and evaluation methods.	SC5	Course Level	SCQF 11
Prepare to learn from others and problem solve.	SC5	Course Level	SCQF 11
Take responsibility for self-managing time and workload scheduling flexibly.	SC5	Course Level	SCQF 11
Appraise the ethical, social, professional and societal issues of an advanced computing problem/solution, and effectively address any such issues.	SC5	Course Level	SCQF 11
Criticise problems and solutions and their alternatives.	SC5	Course Level	SCQF 11

Programme Outcome Types Reference			
#	SCQF Characteristic	Soft Skill	Sub Soft Skill
SC1	Knowledge and understanding	Course Level	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	Course Level	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10

			SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	Course Level	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	Course Level	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	Course Level	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Course Schedules

Stage 1 / Semester 1

Core	
Code	Name
0734 - Electives 1	Advanced Computing - Electives Group 1
0734 - Electives 1	Advanced Computing - Electives Group 1
CMM542	Human Factors in Security
CMM542	Human Factors in Security
CMM542	Human Factors in Security
CMM542	Human Factors in Security
CMM304	Knowledge Modeling and Reasoning
CMM304	Knowledge Modeling and Reasoning

CEM012	Scalable Software
CEM012	Scalable Software
CEM012	Scalable Software
CEM012	Scalable Software

Stage 1 / Semester 2

Core	
Code	Name
CMM547	AI and Data Governance
CMM547	AI and Data Governance
CMM547	AI and Data Governance
CMM547	AI and Data Governance
CMM547	AI and Data Governance
0734 - Electives 2	Advanced Computing - Electives Group 2
0734 - Electives 2	Advanced Computing - Electives Group 2
CMM534	Big Data Analytics and Visualisation
CMM534	Big Data Analytics and Visualisation
CMM534	Big Data Analytics and Visualisation
CMM534	Big Data Analytics and Visualisation
CMM523	Database and Web Security
CMM523	Database and Web Security

Stage 1 / Semester 3

Core	
Code	Name
0734 - Electives 1	Advanced Computing - Electives Group 1
0734 - Electives 1	Advanced Computing - Electives Group 1
CMM304	Knowledge Modeling and Reasoning

CMM304	Knowledge Modeling and Reasoning
CEM100	MSc Project
CEM100	MSc Project
CEM104	Short Postgraduate Placement
CEM104	Short Postgraduate Placement
CEM105	Year-long Postgraduate Placement
CEM105	Year-long Postgraduate Placement

Stage 1 / Semester 4

Core	
Code	Name
0734 - Electives 2	Advanced Computing - Electives Group 2
0734 - Electives 2	Advanced Computing - Electives Group 2
CMM523	Database and Web Security
CMM523	Database and Web Security

Stage 1 / Semester 5

Core	
Code	Name
CEM100	MSc Project
CEM100	MSc Project
CEM104	Short Postgraduate Placement
CEM104	Short Postgraduate Placement
CEM105	Year-long Postgraduate Placement
CEM105	Year-long Postgraduate Placement

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date

Aberdeen, United Kingdom	On-Campus	Full Time	
Aberdeen, United Kingdom	Online	Full Time	
Aberdeen, United Kingdom	On-Campus	Part Time	
Aberdeen, United Kingdom	Online	Part Time	

Sources of Further Information

No sources of further information recorded
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